

International Islamic University Chittagong
Department of Electrical and Electronic Engineering
Complex Engineering Problem

Course Title: Circuit Simulation Sessional
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Problem:

A hybrid analog circuit is shown conceptually below and must be analyzed using circuit-simulation techniques.

The circuit consists of:

A DC voltage source.

A sinusoidal AC source connected in series with the DC source.

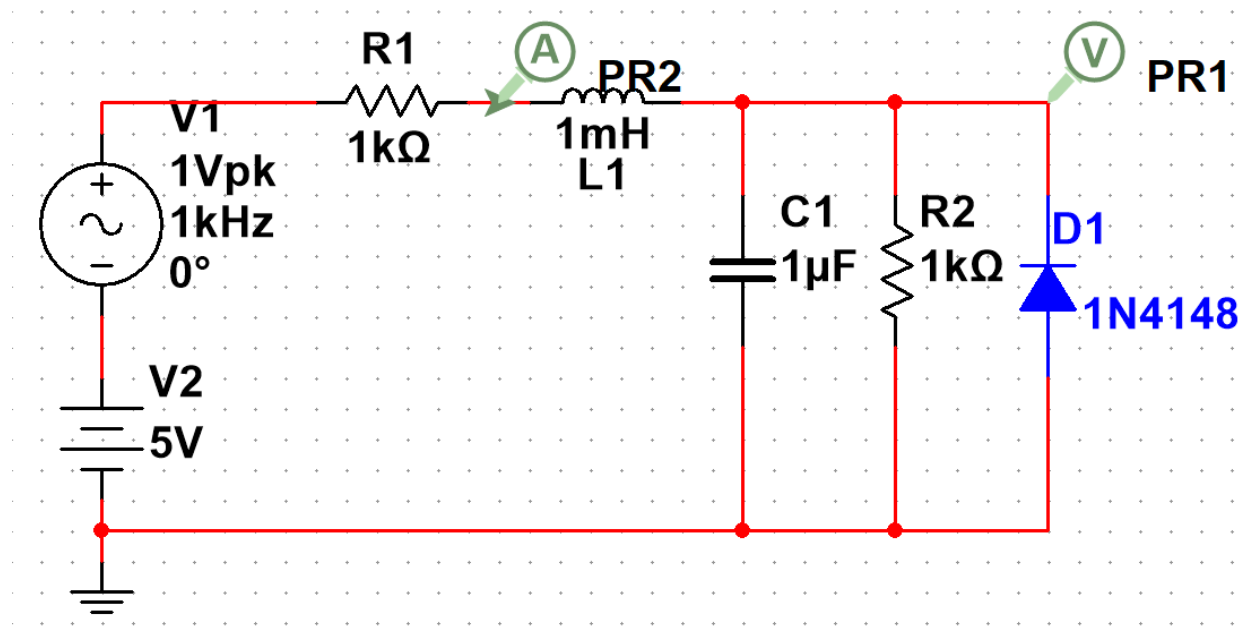
A series RLC branch:

A diode (1N4148) connected in parallel with the capacitor (anode at ground).

A load resistor connected across the capacitor.

Assume ideal wires and standard diode parameters unless otherwise specified.

Ans: NI Multisim diagram

**Task a. DC Operating Point Analysis**

Determine the DC voltage across the capacitor and the current through the diode at steady state. Clearly state all assumptions used in the simulation Task

Ans:

Results:

- **Capacitor Voltage (V_C):** 2.5V
- **Diode Current (I_{D1}):** 0A (Reverse Biased)

Assumptions & Analysis:

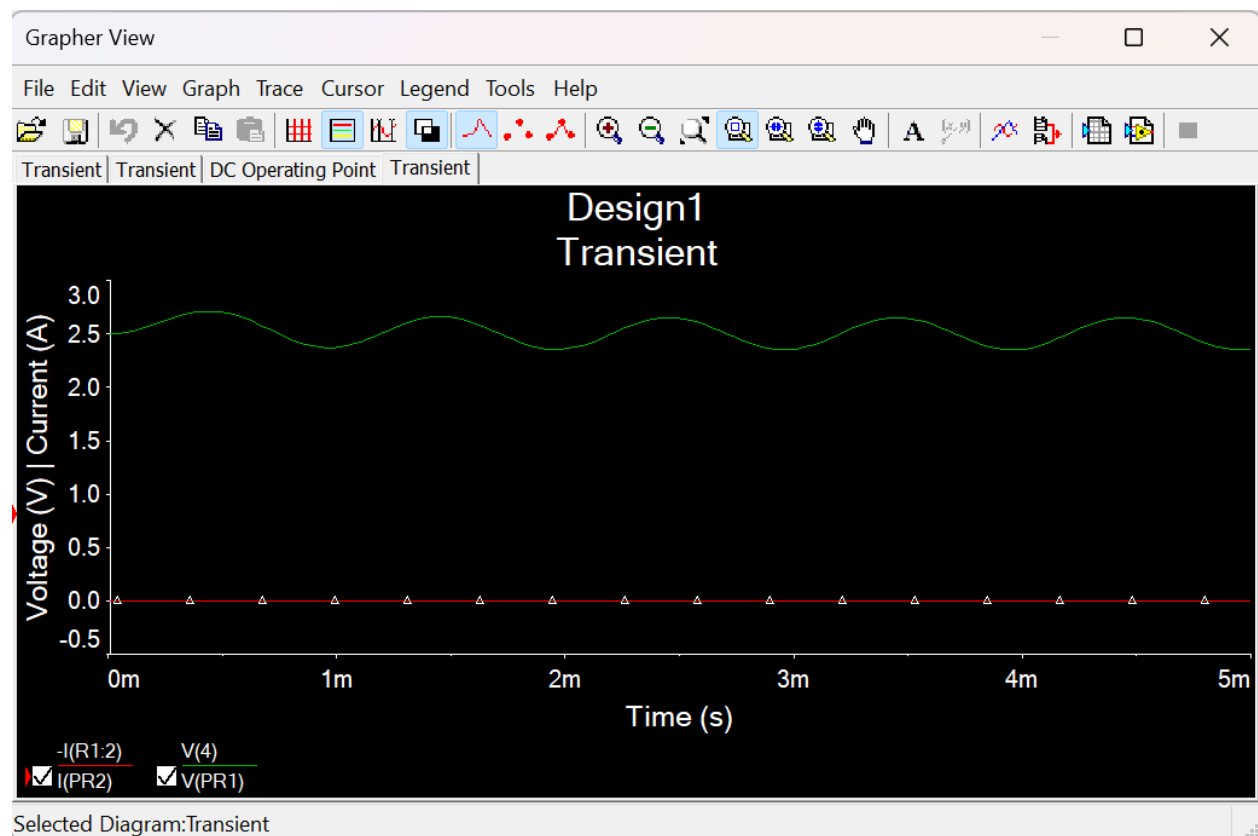
In DC steady state, the inductor acts as a short circuit and the capacitor as an open circuit. The voltage is determined by the resistive divider formed by R_1 and R_2

$$V_{out} = 5V \times \frac{1k\Omega}{1k\Omega + 1k\Omega} = 2.5V$$

Since the cathode is at 2.5V and the anode is grounded, the diode is reverse-biased and does not conduct.

Task b. Transient Response

Using transient analysis, simulate the circuit for Plot:



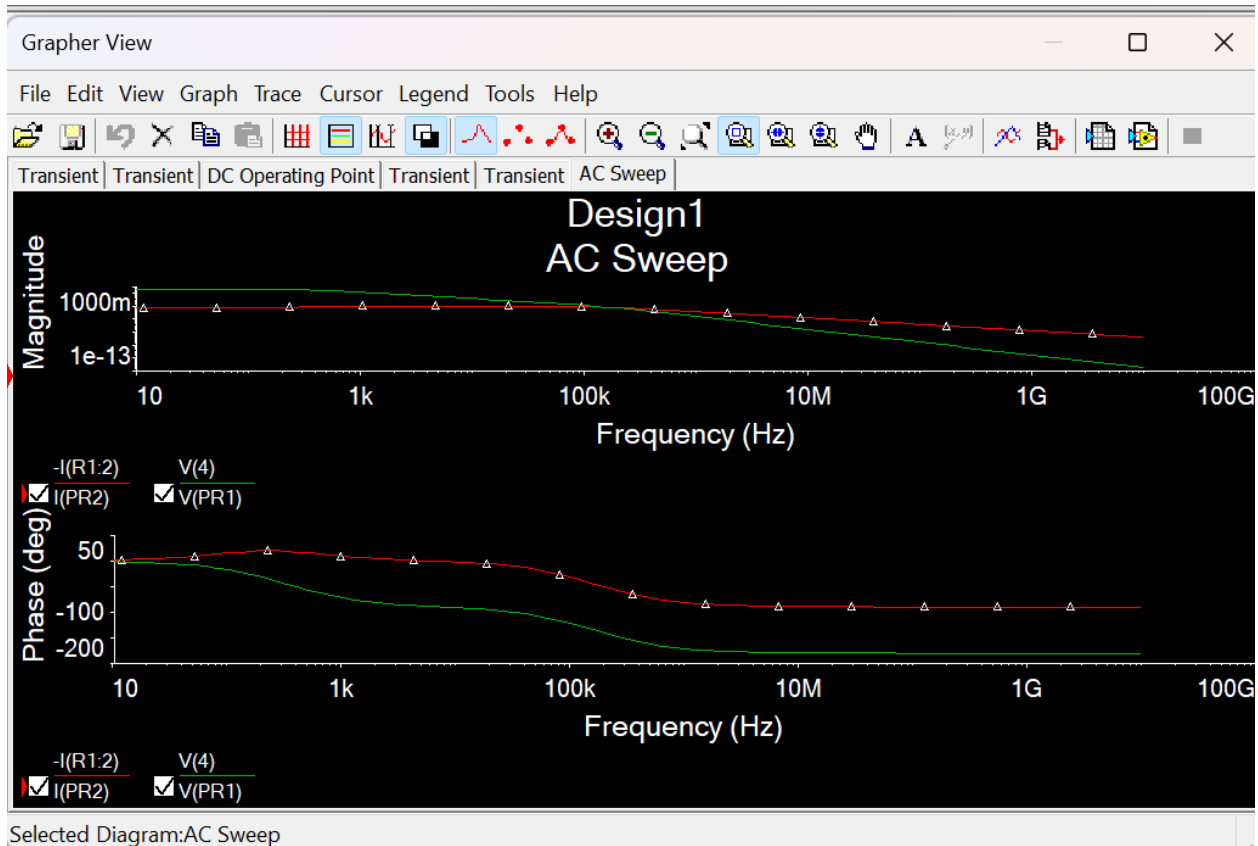
Observations:

- The **capacitor voltage** oscillates as a sine wave centered at the **2.5 V** DC offset.
- The **inductor current** oscillates around a **2.5 mA** DC level.

Effect of the Diode:

The diode has **no effect** on the transient response in this configuration. Because the 1 V_{pk} AC signal rides on a 2.5 V DC pedestal, the voltage never drops below the diode's forward-bias threshold 0.7 V, keeping the diode "OFF" at all times.

Task c. AC Small-Signal Analysis



Resonant Frequency Identification:

Using the simulation cursors, the resonant frequency was found to be approximately 1.59 kHz

Mathematical Proof:

$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10\text{mH} \times 1\mu\text{F}}} \approx 1591.5\text{Hz}$$

At resonance, the output magnitude is at its peak because the reactances of L₁ and C₁ cancel, leaving a purely resistive circuit.

$$V_{out(peak)} = 1\text{V} \times \frac{1000}{1000 + 1000} = 0.5\text{V}$$

$$V_{in(peak)}: 1\text{V}$$

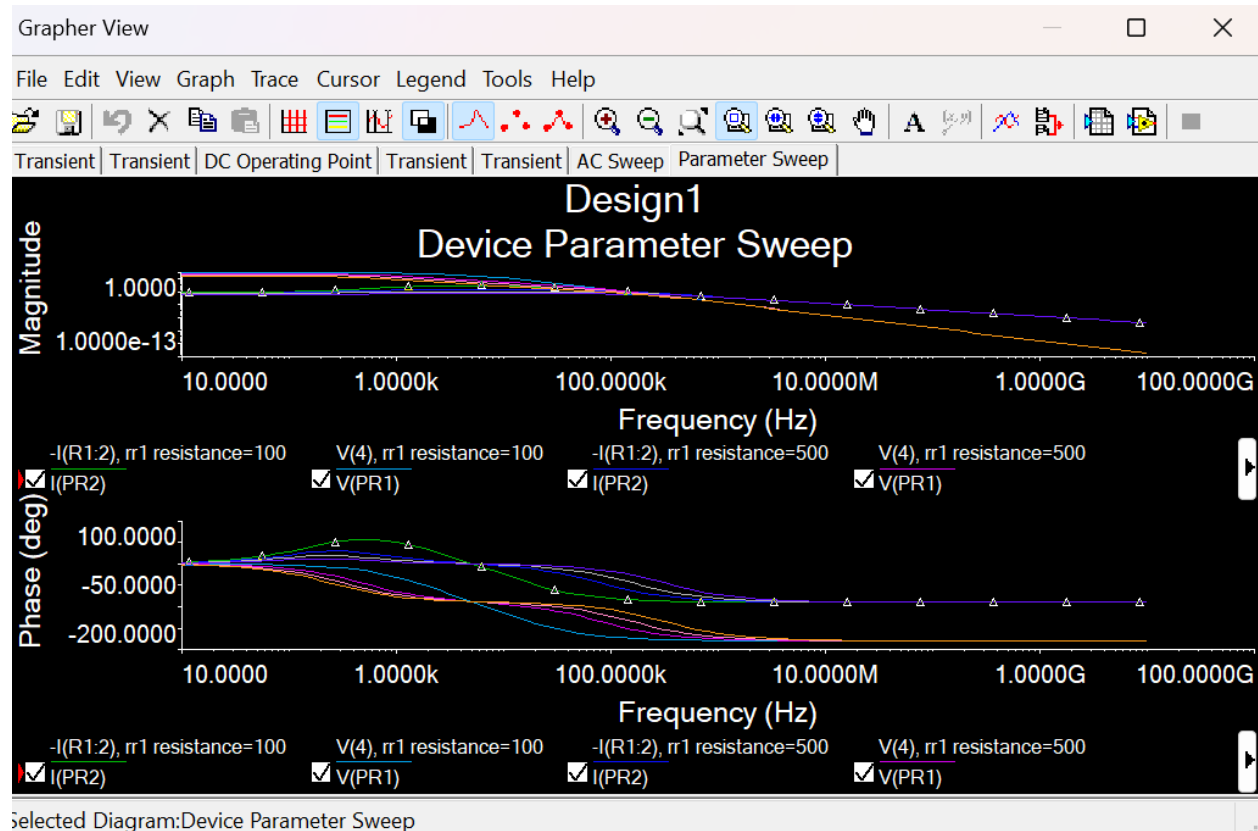
$$R1: 1\text{k}\Omega$$

$$R2: 1\text{k}\Omega$$

Task d. Parametric Study

By varying from to, analyze and explain how the damping factor and bandwidth of the circuit change. Support your answer using simulation results.

Ans:



Analysis: By varying R_1 (100 Ω , 500 Ω , 1k Ω , 2k Ω) the following was observed:

Damping Factor: As R_1 increases, the **damping factor increases**, causing the resonant peak to flatten and decrease in magnitude.

$$\zeta = \frac{R_{total}}{2} \sqrt{\frac{C}{L}} \quad \text{and} \quad \sqrt{\frac{C}{L}} = \sqrt{\frac{1 \times 10^{-6}}{0.01}} = 0.01$$

R1 Value	R _{total} (R1+R2)	Damping Factor (ζ)
100 Ω	1100 Ω	0.55 (Underdamped)
500 Ω	1500 Ω	0.75 (Underdamped)
1000 Ω	2000 Ω	1.00 (Critically Damped)
2000 Ω	3000 Ω	1.50 (Overdamped)

Bandwidth: As R1 increases, the **bandwidth increases**. The circuit becomes less selective, spreading the energy across a wider frequency range.

$$BW = \frac{R_{total}}{2\pi L}$$

Using $2\pi L=0.06283$, the values are:

R1 Value	R _{total}	Bandwidth (BW)
100Ω	1100Ω	17.51 kHz
500Ω	1500Ω	23.87 kHz
1000Ω	2000Ω	31.83 kHz
2000Ω	3000Ω	47.75 kHz

Task e. Simulation Accuracy

Mention two possible sources of error in circuit simulation compared to real hardware implementation and briefly explain their impact.

Ans:

Two primary sources of error between simulation and hardware are:

1. **Ideal Inductor Models:** Multisim assumes L1 has no internal resistance. Real inductors have **DC Resistance (DCR)** which would result in higher damping and lower output voltage in a physical circuit.
2. **Parasitic Capacitance:** Physical circuit boards and components have **stray capacitances** not present in the simulation. These can cause the actual resonant frequency to shift slightly from the theoretical 1.59 kHz.